

Pichai changed the way Google looked at future opportunities for disruptive growth and impact.

DeepMind was something that Pichai had in mind. He had seen all the work done by them in such a short time and decided to acquire the company back in 2014. This allowed Google to lead with a lot of innovative AI products and services, backed by research done by the AI unit run by Nobel laureate Demis Hassabis.

Google also shifted their focus to building the Tensor Processing Unit (TPU), and it was first deployed internally back in 2015. These chips were designed specifically with one thing in mind, and that was to accelerate “TensorFlow”, which is Google’s open-source machine learning library.

Not only that, but looking back at 2016 now, it can be seen that Google heavily invested a lot of their resources in data centres. They poured in tens of billions of dollars to set up data centres all over the globe. While most might have called this decision a big gamble at the time, given today’s day and age, it is safe to say this was a crucial step they took early on.

How Google embedded AI

Whether you like it or hate it, it’s safe to say that Google really changed the way we interact with AI. Sure, they have an AI chatbot in the form of Gemini; however, even before the launch of the chatbot, they quietly integrated numerous AI features into their most popular products.

Firstly, let’s start with the most popular one, Google Search. The introduction of RankBrain, back in 2015, was the stepping stone for how we interact with search today. It allowed Search to give relevant results even if the user didn’t type all the exact words required for the search. It worked by understanding all the content that is related to other words and concepts.

Next up, Gmail was the next popular product, where Google decided

to implement their AI systems. Many didn’t even realise it, but features like smart reply and smart compose were what brought AI to Gmail. Both these features used AI to predict what the next word you would use.

What’s interesting is that while it was a convenient feature for many, on the other hand, for Google, it worked as a live training model for their AI. Feedback is critical when testing new features on any AI. By integrating the features into Gmail, Google essentially got feedback for its AI model every time someone used the feature.

Something similar was seen with Google Photos as well. The AI used in the Photos app was pretty huge, as instead of having to rename a par-



ticular picture or endlessly scroll, the AI model understood every pixel of all the photographs captured.

Not only did it help in finding pictures easily, but it also helped train Google’s AI for the future. After just a few years of introducing AI into Google Photos, they managed to make smart features like Magic Eraser. This was all thanks to all the data they collected, which helped the Magic Eraser understand what to replace objects with.

Finally, AI was even introduced to Google Maps. I’m sure that you look at Google Maps before leaving for any destination. And that’s clearly because of the smart features added to Maps, which make it easy to understand many things about the route.

By implementing the AI into Maps, Google integrated machine learning to analyse historical traffic patterns while also understanding the real-

time data that it already gets. All in all, this smart combination helped in predicting traffic before it even happened.

Though the integration didn’t stop at user utilities, it also made Google rethink their revenue streams. For example, with Google Ads, they integrated AI in a way that would allow them to move from simple keyword matching to Smart Bidding. This feature uses ML to analyse millions of signals in milliseconds to set the perfect bid for every single auction.

While on YouTube, AI algorithms shifted from merely suggesting videos to predicting viewer intent, allowing for targeted advertising that feels native to the user experience. By embedding AI into AdSense, Google automated the “Quality Score” system. Rewarding relevance and ensuring that every ad attached to a video is aligning with a user’s actions.

It is safe to say that by implementing all these features into their popular services, Google really played it smart. On one hand, it was a big bet, as AI was still new. But on the other hand, it also helped them get the most amount of feedback that they could have gotten. Not to forget, all this also helped them in training their AI, which is now known as Gemini.

Google's rights (and wrongs)

Everyone makes mistakes, and that’s how they learn. The same can be said for Google. While they have perfected many things, which have given them a head start in the AI race, it is viable to say that they’re still not in first place.

The Bard controversy serves as an example. In the rush to respond to the sudden public hype for generative AI, Google’s rollout faced criticism. There were many factual errors during their live demos that it even led to many questioning Google’s future in AI. **d**

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